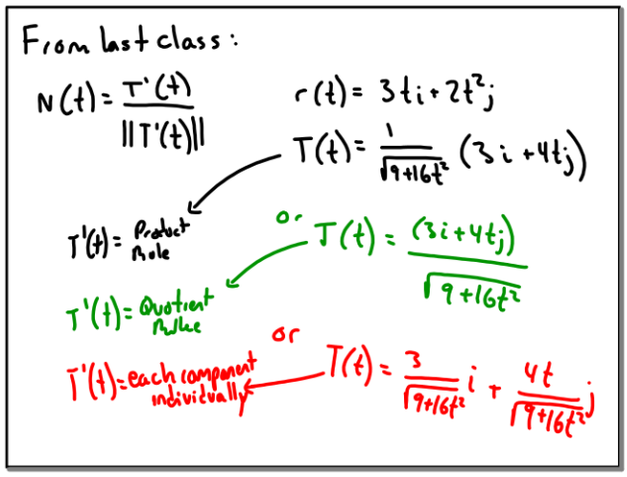
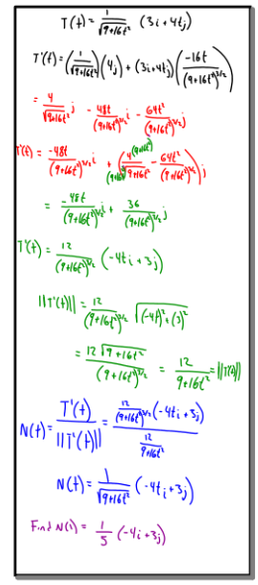
Ameet Rao's Math Notes

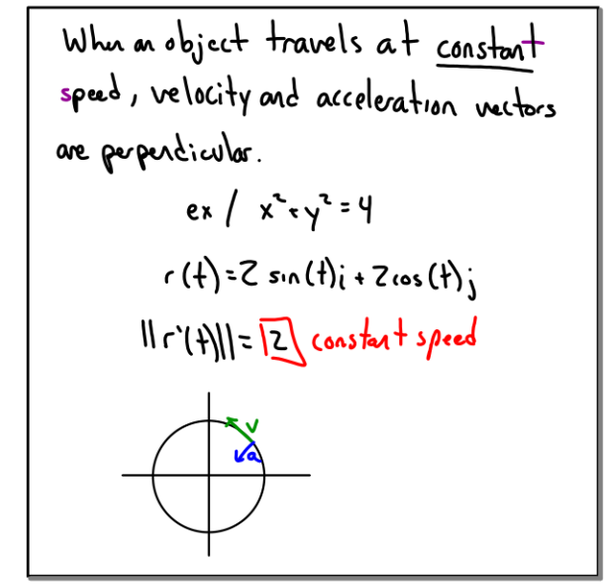
Principal Unit Normal, Constant Speed, and Tangential and Normal Acceleration



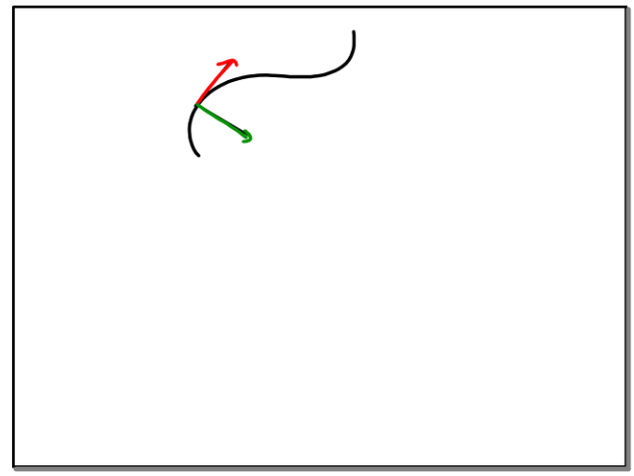
To differentiate T(t) = (3i + 4t j)/√(9 + 16t²), use the product rule, the quotient rule, or split it into 3/√(9 + 16t²) i + 4t/√(9 + 16t²) j.



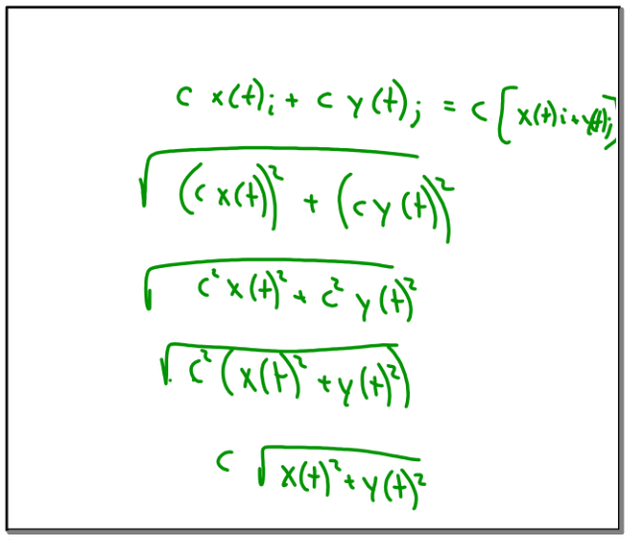
Worked through: T′(t) = 12(−4t i + 3j)/(9 + 16t²)^(3/2) and ‖T′‖ = 12/(9 + 16t²), so N(t) = (−4t i + 3j)/√(9 + 16t²) and N(1) = (−4i + 3j)/5.



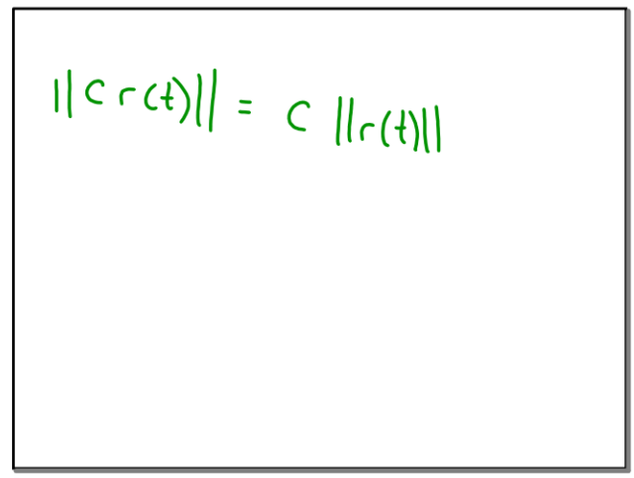
At constant speed, velocity and acceleration are perpendicular. On x² + y² = 4, r(t) = 2sin t i + 2cos t j has ‖r′‖ = 2, so a points to the center.



Sketch: the velocity arrow is tangent to the path, while the acceleration arrow points toward the inside of the turn.

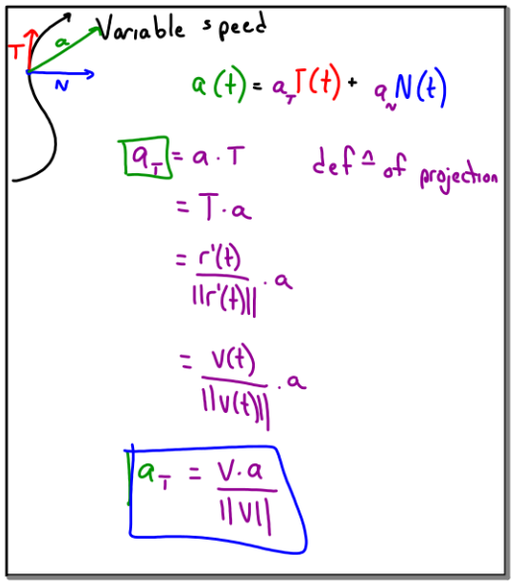


A scalar factors out of a length: ‖c r(t)‖ = √((cx)² + (cy)²) = √(c²(x² + y²)) = |c|·√(x² + y²).

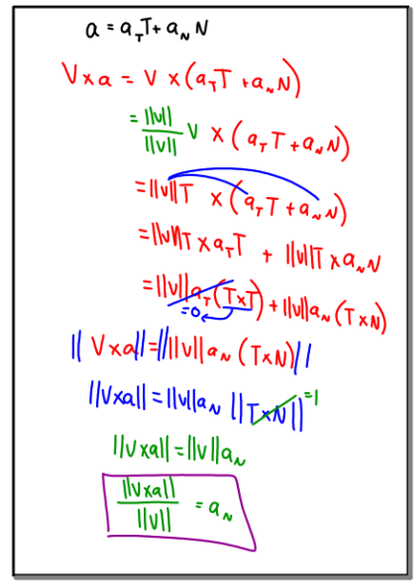


So ‖c r(t)‖ = |c| ‖r(t)‖. Strictly it is |c|, since a length can't be negative.

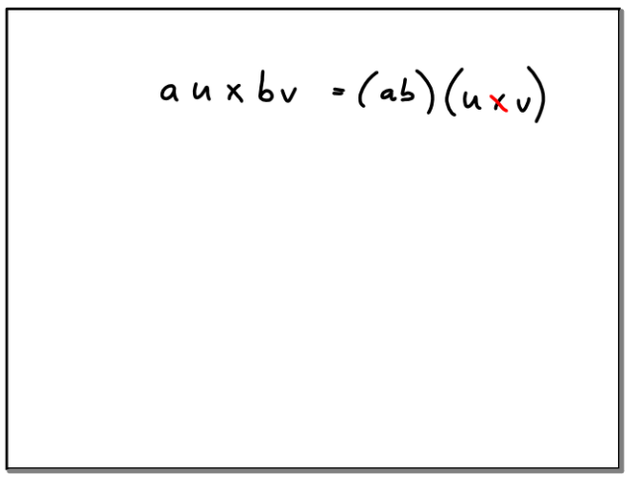
Produced in 2026



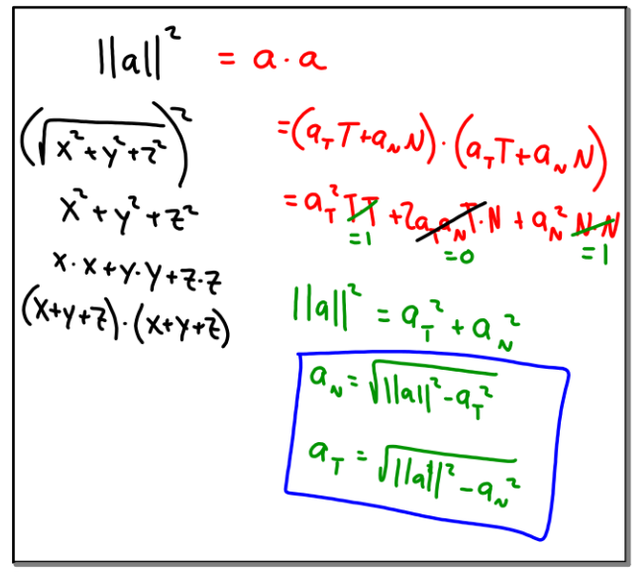
With variable speed, a = a_T T + a_N N. The tangential part is the projection of a onto T: a_T = a·T = (v·a)/‖v‖.



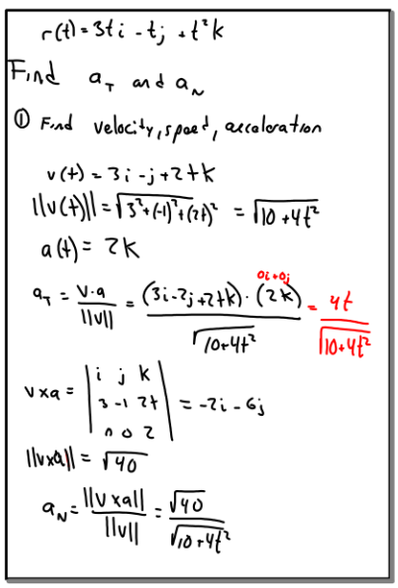
Cross v with a = a_T T + a_N N. Since T × T = 0 and ‖T × N‖ = 1, ‖v × a‖ = ‖v‖ a_N, giving a_N = ‖v × a‖/‖v‖.



Scalars come out of a cross product: (a u) × (b v) = ab (u × v). That is the step that pulls ‖v‖ and a_T out in the derivation above.



T and N are perpendicular unit vectors, so ‖a‖² = a·a = a_T² + a_N². Either component follows from the other: a_N = √(‖a‖² − a_T²).



Example r = 3t i − t j + t² k: v = 3i − j + 2t k, ‖v‖ = √(10 + 4t²), a = 2k. So a_T = 4t/√(10 + 4t²); v × a = −2i − 6j gives a_N = √40/√(10 + 4t²).

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